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CRITICAL REVIEW

Advances and limitations in machine learning approaches applied to remaining useful life predictions: a critical review

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Abstract

Predictive maintenance (PdM) is critical to ensure optimal operating efficiency and minimize costly failures of industrial machinery. The PdM leverages a machine learning (ML) method to predict remaining useful life (RUL) for implementing minimal-cost and reliable maintenance. RUL prediction involves multiple steps, such as data collection, data pre-processing, and RUL estimation, each of which incorporates various methods. This study conducted a critical review of RUL estimation and data pre-processing specifically for turbofan engines and bearings, categorizing existing models to offer a high-level perspective on RUL prediction. This review demonstrates that the indirect mapping method exhibits outstanding prediction accuracy compared to the direct mapping method. Moreover, it highlights that autoencoder techniques and their variants demonstrate commendable performance in extracting features from turbofan engines and bearing datasets. Furthermore, the paper proposes potential areas for future research to improve RUL prediction in this domain.

Keywords RUL prediction · Turbofan engine · Bearing · Critical review

Abbreviations

PM	Preventive maintenance
PdM	Predictive maintenance
RUL	Remaining useful life
ML	Machine learning
TL	Transfer learning
NASA	National Aeronautics and Space Administration
C-MAPSS	Commercial Modular Aero-Propulsion System Simulation
N-CMAPSS	New C-MAPSS
mRMR	Max-Relevance and Min-Redundancy
PHM	Prognostic Health Management
IMS	Intelligent Maintenance System
WAMP	Wilson amplitude
FOT	Fault occurrence time
SVM	Support vector machine
HHT	Hilbert–Huang Transform
HIs	Health indicators
RF	Random forest
ELM	Extreme learning machine
CNN	Convolutional neural network
LSTM	Long short-term memory
HMM	Hidden Markov Model
PCA	Principal component analysis
DBN	Deep belief network
RNN	Recurrent neural network
MAE	Mean absolute error
CRA	Cumulative relative accuracy
RMSE	Root mean square error
LRT	Linear rectification technique
FFT	Fast Fourier Transform
TCN	Temporal convolution network

1 Introduction

The annual maintenance cost for machinery is average 30% of the operating cost, covering factors like the fuel, maintenance, tires, repairs, as reported by Shukri et al. [1]. Maintenance is crucial because it not only ensures the optimal efficiency of the machinery and equipment required for production processes but also minimizes expensive failures. The three commonly used maintenance strategies in industry are reactive, preventive, and predictive maintenance [2], and these are implemented at different operational phases, as

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illustrated in Fig. 1. Reactive maintenance involves repairing machinery after it fails; this strategy is predominantly used in early industrial stages. However, it causes delays in maintenance and fails to account for losses resulting from unforeseen faults. By contrast, preventive maintenance (PM) involves the periodic execution of maintenance tasks according to planned schedules that are based on the experience of experts [3]. A planned schedule characterized by excessively short intervals between maintenance activities may lead to high maintenance costs, while a planned schedule with lengthy time allowances between tasks may result in unanticipated failures [4]. Meanwhile, predictive maintenance (PdM) relies on data analysis tools to schedule maintenance using remaining useful life (RUL) predictions rather than solely relying on expert experience. Hence, by leveraging data analysis for real-time machine-status prognostics, PdM maximizes the equipment service life and minimizes maintenance costs. As a result, using PdM rather than PM can reduce maintenance costs by 20–30% and machine downtime by 20–50%. Consequently, PdM has emerged as a promising maintenance strategy and it is rapidly expanding.

PdM requires a high degree of RUL prediction accuracy to obtain a cost-effective and reliable maintenance. Using machine learning (ML) as data analysis tools has proven to be an effective approach for learning and describing the nonlinear and complex degradation processes required to predict the RUL [5, 6]. RUL predictions that are based on ML methods include two steps, as shown in Fig. 2: data pre-processing and RUL estimation. Data pre-processing is a technique used to eliminate redundant and useless data as well as to perform normalization to reduce the influence of irrelevant information on the accuracy of RUL predictions. RUL estimation involves the process of learning the degradation patterns and subsequently predicting the RUL. It can be achieved through either direct or indirect mapping methods [7]. Direct mapping methods utilize ML techniques to establish optimal mapping function between pre-processed signals and RUL values. Conversely, indirect mapping methods focus on identifying mapping function between features extracted from the pre-processed data and RUL values. While multiple models exist for each step of RUL prediction, a comprehensive review of these models, encompassing pre-processing and RUL estimation steps, is crucial for PdM. Such reviews offer insights to enhance ML model selection for more accurate predictions and future research directions.

Lei et al. [8] and Zhang et al. [9] evaluated different methodologies (physical, statistical, deep learning, and hybrid) for RUL prediction, investigating models like Paris-Erdogan and Artificial Neural Network (ANN). Chen et al. [10] delved into cutting-edge transfer learning (TL)-based RUL predictive solutions. Huiqin Li et al. [11] comprehensively reviewed the Physics-informed data-driven remaining useful life prediction methods. Notably, these studies overlooked the intricacies of the RUL prediction process. To tackle these omissions, Ferreira et al. [5] outlined 14 challenges distributed across macro and micro steps within RUL prediction using existing datasets, summarizing how ML methods address these challenges. Lei et al. [12] systematically surveyed machinery prognostics, specifically in data acquisition, HI construction, HS division, and RUL prediction processes. Despite these, a deficiency was noted in their critical analysis and classification of ML models for RUL prediction, including a lack of quantitative evaluation. Although Kim et al. [13–16] and Vollert et al. [17] quantitatively examined a range of data-driven RUL prediction methods, their primary focus on individual datasets. Kim et al. [13–16] focused on RUL prediction of Li-ion battery, and Vollert et al. [17] concentrated on turbofan engines.

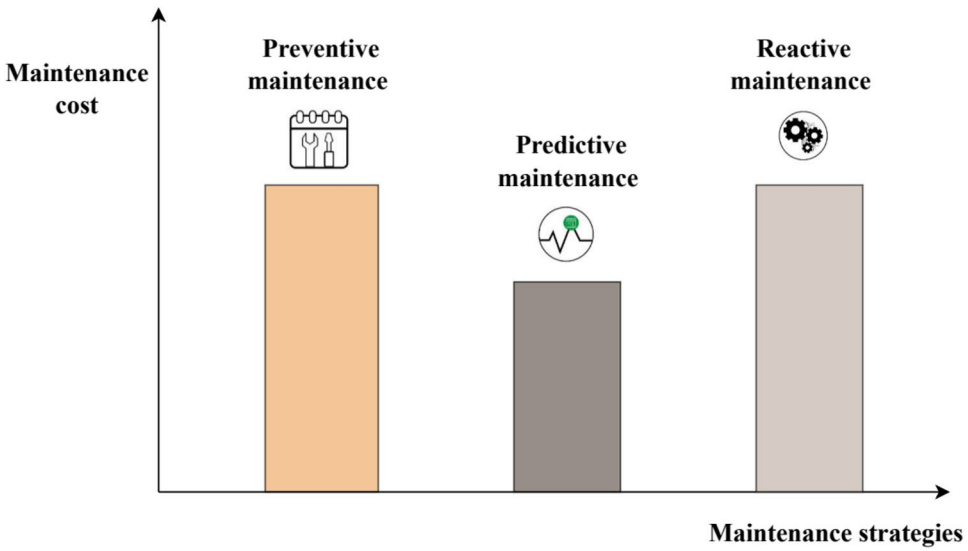


Fig. 1 Comparison of the maintenance costs associated with reactive, preventive, and predictive maintenance

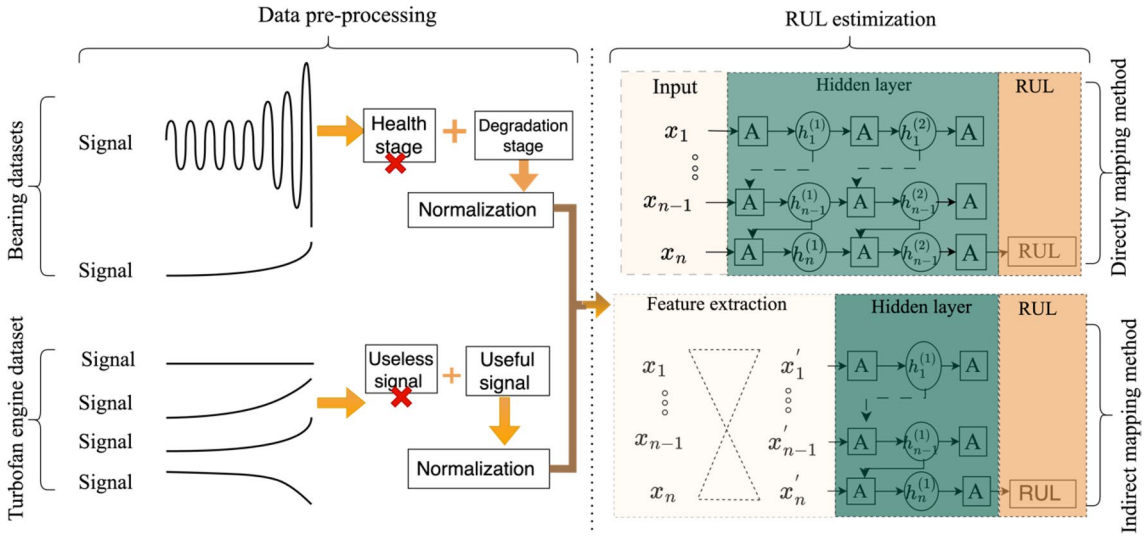


Fig. 2 Turbofan engine and bearing RUL prediction methods based on machine learning

Consequently, a comprehensive critical review of RUL prediction techniques throughout the entire process is essential for PdM advancement across various datasets.

In this study, turbofan engines and bearings [5]—commonly utilized research subjects for RUL prediction—were selected to assess and compare the prediction accuracies of RUL estimation along with data pre-processing techniques. Furthermore, for comparison purposes, the RUL estimation was classified as either direct or indirect mapping methods, then they were further categorized based on their structural similarities. The remainder of this paper is divided into three sections. In Section 2, several datasets used for RUL prediction are introduced and data pre-processing techniques for bearing and turbofan engine datasets are compared. Section 3 presents a critical review of RUL estimation as well as a discussion of the model performance assessment and weighting, which was based on the prediction accuracy. Finally, Section 4 provides a formal conclusion for this work and outlines directions for future research.

2 Data pre-processing

The development of data pre-processing techniques has great potential for enhancing data quality and improving RUL estimations because they eliminate redundant and irrelevant data. To comprehensively address the data pre-processing techniques, a set of 15 high-quality datasets, which includes turbofan engine and bearing datasets, was collected from the U.S. National Aeronautics and Space Administration (NASA), the IEEE Reliability Society, XJTU-SY, and the University of Cincinnati are the subjects in this paper [5]. The following sections introduce the datasets used during this study and present a critical review of existing data pre-processing techniques.

2.1 Turbofan engines

2.1.1 Dataset descriptions

With the goal of preventing approximately 50% of aircraft failures [5], NASA published the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset and collaborated with ETH Zurich to publish the New C-MAPSS (N-CMAPSS) dataset, thereby improving RUL predictions for turbofan engines. The C-MAPSS dataset consists of simulated run-to-failure degradation data from turbofan jet engines generated using the C-MAPSS dynamic model [18]. Figure 3 presents a simplified diagram of the C-MAPSS dynamic model. In this

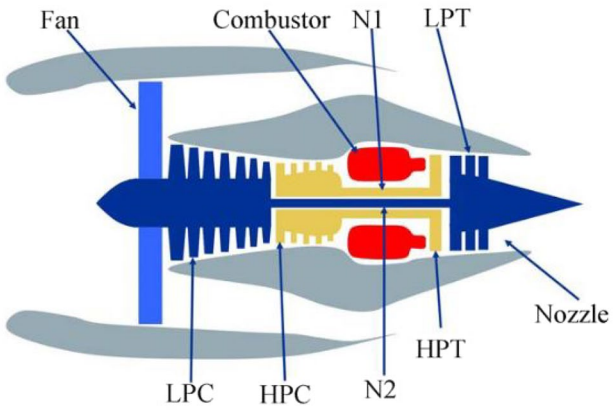


Fig. 3 Simplified diagram of the engine in the C-MAPSS model [18]

model, there are 21 sensors on the turbofan engine that collect temperature, pressure, and speed signals to monitor the condition or degradation status [19]. In addition, the C-MAPSS dataset is divided into four subsets, FD001, FD002, FD003, and FD004, according to the operating conditions and fault modes, as described in Table 1. Moreover, each subset contains three operating settings and 21 signals, including temperature, pressure, and speed signals, to monitor the condition or degradation status, as shown in Fig. 4.

In addition, NASA collaborated with ETH Zurich to publish a new dataset of turbofan engine run-to-failure trajectories named the new C-MAPSS (N-CMAPSS) dataset. This dataset aims to closely replicate real-world scenarios. Comprising 31 sensor signals, it includes 13 signals derived from actual flight data and 18 signals simulated to enhance its representation of authentic operating conditions [20]. Figure 5 presents a simplified diagram of the N-CMAPSS model.

2.1.2 Data pre-processing techniques

After online data acquisition has been completed, pre-processing techniques are vital to the selection of reliable attributes and the removal of data that do not significantly impact the RUL. The C-MAPSS and N-CMAPSS turbofan engine datasets have 21 and 31 sensor signals, respectively. However, some sensors do not contain enough degradation information to be useful for training ML models for RUL estimation. Therefore, ML methods are used to remove insignificant information so that signals relevant to the degradation process can be extracted.

Using the C-MAPSS dataset above, Zhang et al. [21] found that only 14 of the 21 signals outlined in Fig. 4 are relevant to the degradation; therefore, the irregular and unchanged signals were eliminated using a virtual method. Moreover, trend analysis, Lasso regression, and Max-Relevance and Min-Redundancy (mRMR) methods [22–24] were used by Zhang et al. to determine that only 14 out of 21 signals are relevant to the degradation for RUL estimation. Similarly, Koutroulis et al. proposed the nonlinear Granger causality method to select 10 out of the 14 measurement signals from the N-CMAPSS dataset [25]. Further development of pre-processing techniques is required for the N-CMAPSS dataset since it is still in its early stages of utilization.

2.2 Bearings

Bearings typically have long lifespans when they are used under normal operating conditions. The IEEE Reliability Society [26], XJTU-SY [27], and the University of Cincinnati [28] proposed the addition of a radial force to the bearing support to expedite the bearing degradation process.

2.3 Dataset descriptions

2.3.1 IEEE 2012 Prognostic Health Management (PHM) bearing dataset

The IEEE 2012 PHM bearing dataset is a collection of experimental data from 17 ball bearings provided by the IEEE Reliability Society. The dataset consists of sets of six and 11 bearings used for training and testing purposes, respectively. The data were collected from an accelerated degradation test rig, shown in Fig. 6, in which four bearings were simultaneously tested until failure under constant radial load and rotational speed values. The bearings were tested for three distinct operating conditions:

- Condition 1: A rotational speed of 1,800 r/min and a radial force of 4,000 N,
- Condition 2: A rotational speed of 1,650 r/min and a radial force of 4,200 N,
- Condition 3: A rotational speed of 1,500 r/min and a radial force of 5,000 N.

The experimental platform for the acceleration bearings included three parts: a rotary mechanism, a radial force generator, and data acquisition equipment. An AC motor applied rotational energy to the shaft, while a force actuator generated a force that acted on the bearing. At the measurement point, vibration signals along the vertical and horizontal axes and a temperature signal were collected using temperature and vibration sensors. The collected data were then divided into training and test datasets, as shown in Table 2.

2.3.2 XJTU-SY bearing dataset

Lei et al. cooperated with Zhejiang Changxing Sumyouung Technology Co. to establish a similar experimental platform to test accelerated bearing degradation, which is depicted in Fig. 7 [27]. The XJTU-SY bearing data were collected under three distinct operating conditions:

- Condition 1: A rotational speed of 2,100 r/min and a radial force of 12,000 N,
- Condition 2: A rotational speed of 2,250 r/min and a radial force of 11,000 N,
- Condition 3: A rotational speed of 2,400 r/min and a radial force of 10,000 N.

While the experimental platform used for the XJTU-SY bearing dataset applied significantly higher radial forces than

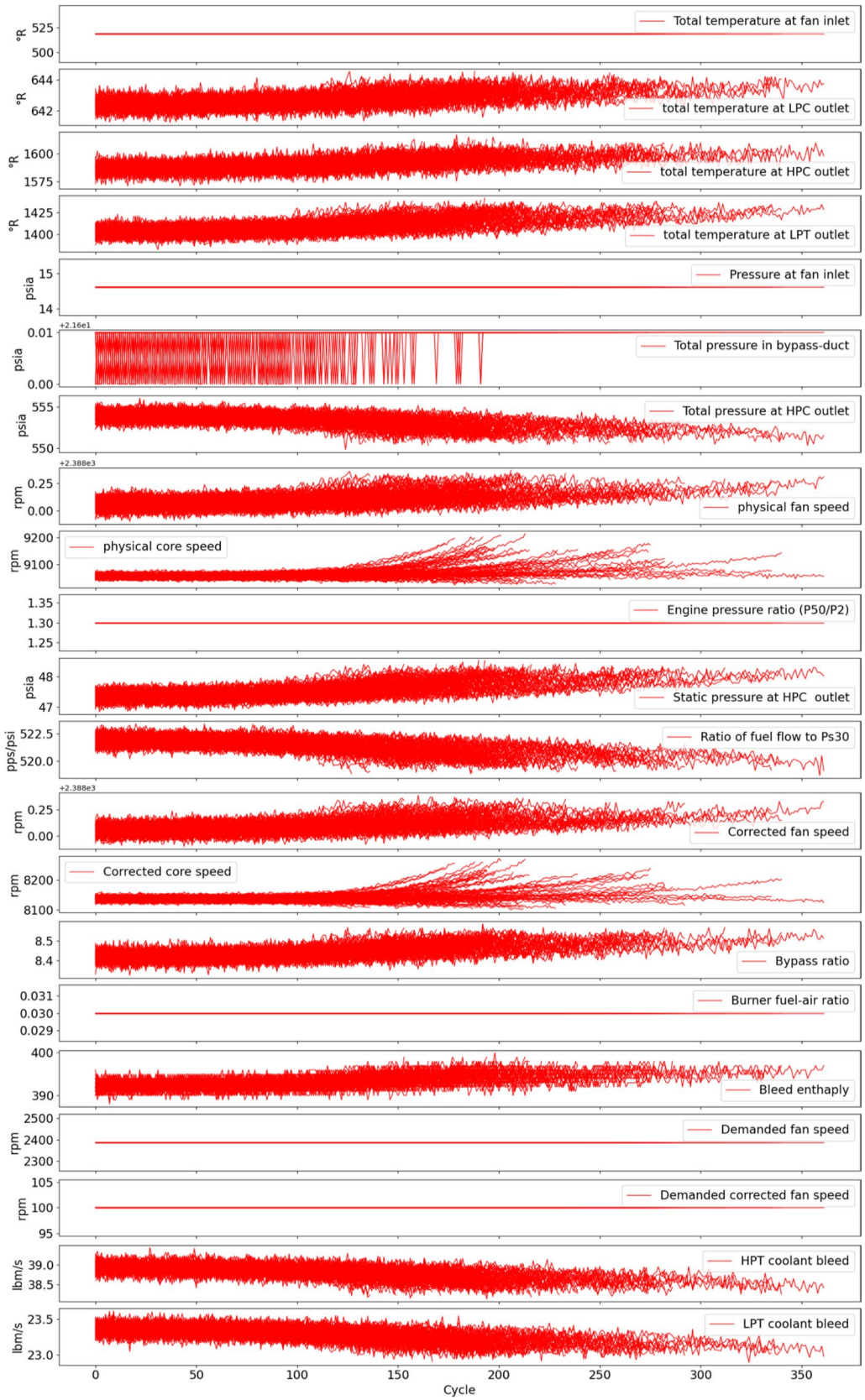


Fig. 4 Data visualization for the FD001 dataset.

Table 1 Descriptions of the C-MAPSS datasets

	FD001	FD002	FD003	FD004
Number of engines in the training set	100	260	100	249
Number of engines in the test set	100	259	100	248
Number of training samples	20,631	53,579	24,720	61,249
Number of test samples	13,096	33,991	16,596	41,214
Operating conditions	1	6	1	6
Fault modes	1	1	2	2

those applied during the collection of the IEEE 2012 PHM bearing dataset, similar bearing working life values were obtained, as indicated in Table 3.

2.3.3 Bearing dataset from the University of Cincinnati

The Intelligent Maintenance System (IMS) at the University of Cincinnati also built a similar experimental platform, as shown in Fig. 8, to investigate the RULs of different bearings [28]. This experimental platform consisted of a motor, a bearing, accelerometers, thermocouples, and a radial force generator. The data generated from this experiment were partitioned into three subsets describing the bearing run-to-failure processes. Each subset contains individual files that include the x- and y-axis vibration and temperature signals for a sampling rate of 20 kHz. However, this dataset was generated for only one operating condition, thereby resulting in low trained-model robustness for RUL prediction.

2.3.4 Data pre-processing techniques

Bearings exhibit an extended period of healthy operation before transitioning into a degraded stage, and it is apparent that data collected during these distinct stages differ significantly. To understand the bearing degradation process well, it is essential to exclude the healthy-stage data from the bearing datasets. Therefore, some pre-processing techniques were proposed to identify and discard the healthy-stage data and keep the degraded-stage data. For instance, Li et al. [29] used the Wilson amplitude (WAMP) feature to automatically detect the fault occurrence time (FOT) and divide the bearing degradation process into two stages: healthy and degraded. However, the FOT is detected solely from the vibration amplitude signal, while other features of the vibration signals are ignored. Furthermore, Wu et al. [30] proposed the support vector machine (SVM) to separate the healthy and degraded stages after extracting time and frequency features of the vibration signals using the Hilbert–Huang Transform (HHT). However, this method is sensitive to noise in the training dataset, which affects the vibration signal classification in complex domains and high-dimensional space.

3 ML models for RUL estimation

After the data acquisition and pre-processing phases, RUL estimation is utilized for RUL prediction. ML models were critically reviewed and assessed during this study to provide high-level insight into the existing knowledge regarding RUL estimation for turbofan engines and bearings. Relevant articles were retrieved using the keywords "turbofan engine," "bearing," and "remaining useful life (RUL)" in the IEEE Xplore, Web of Science, and ScienceDirect databases.

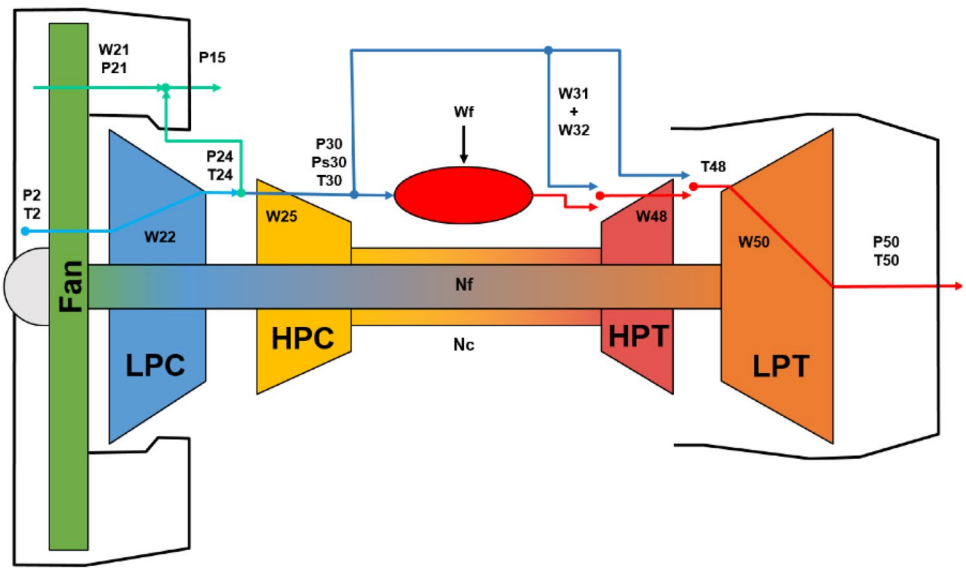


Fig. 5 Simplified diagram of the N-CMAPSS model [20].

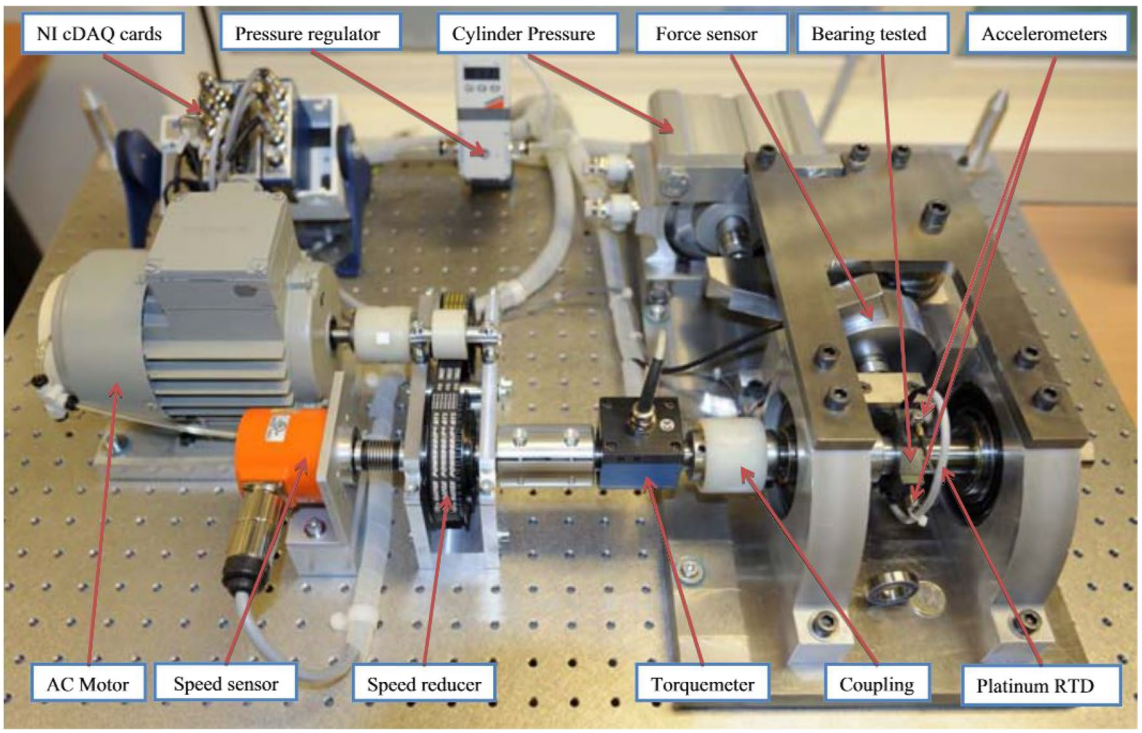


Fig. 6 Photograph of the experimental platform used to investigate accelerated bearing degradation [26].

Table 2 Overview of the IEEE 2012 PHM bearing dataset

Datasets	Operating Conditions		
	Condition 1	**Condition 2**	**Condition 3**
Training dataset	Bearing 1_1	Bearing 2_1	Bearing 3_1
	Bearing 1_2	Bearing 2_2	Bearing 3_2
Test dataset	Bearing 1_3	Bearing 2_3	Bearing 3_3
	Bearing 1_4	Bearing 2_4	
	Bearing 1_5	Bearing 2_5	
	Bearing 1_6	Bearing 2_6	
	Bearing 1_7	Bearing 2_7	

Then, the article selection process was continued by selecting a precise date range (2017–2023) and excluding open-source articles. Finally, 110 relevant and high-quality articles were analyzed. Of these articles, 83 focused on turbofan engine RUL estimation, while the rest were dedicated to bearing RUL estimation. Consequently, the goal for this section was to critically review RUL estimation for turbofan engines and bearings.

3.1 Turbofan engines

Among the 83 articles that discuss turbofan engine RUL estimation, 17 articles discussed the development of direct mapping methods to estimate the RUL, while the remaining 66 articles focused on indirect mapping methods. Direct mapping methods estimate the RUL by constructing a mapping relationship between the pre-processed data and the corresponding RUL values. By contrast, indirect mapping methods involve extracting representative temporal and spatial features, referred to as health indicators (HIs), from the pre-processed data when these features are one-dimensional. Subsequently, the extracted features are fed into predictors to estimate the RUL. Therefore, separate critical reviews and performance assessments were conducted for the direct and indirect mapping methods during this study.

3.1.1 Direct mapping methods

Direct mapping methods find the optimal relationships between the pre-processed data and the corresponding RUL values using ML models [31]. For instance, Mathew et al. [32] proposed a random forest (RF) algorithm to predict the RUL. This RF algorithm combines the outputs of multiple decision trees to make accurate and stable predictions. However, the decision trees have difficulty obtaining the best split when the data features are highly correlated with each other. Furthermore, Berghout et al. [33] proposed an extreme learning machine (ELM) to improve the computational efficiency of RUL estimation by changing the weight of the input layer. However, it is not suitable for turbofan engine datasets because it assumes that the

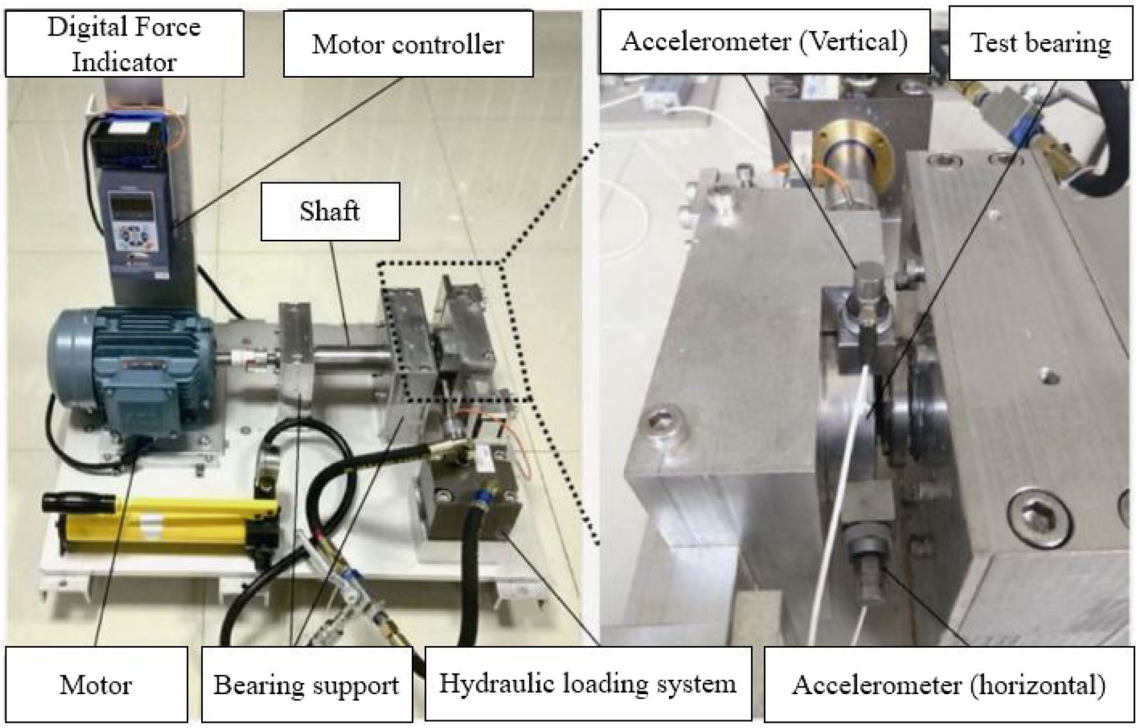


Fig. 7 Accelerated bearing degradation test bench [27].

Table 3 Overview of XJTU-SY bearing dataset

Operating condition	Datasets	Working life	Failure location
1	Bearing 1_1	2 h, 3 min	Outer ring
	Bearing 1_2	2 h, 41 min	Outer ring
	Bearing 1_3	2 h, 38 min	Outer ring
	Bearing 1_4	2 h, 2 min	Cage
	Bearing 1_5	52 min	Inner ring and outer ring
2	Bearing 2_1	8 h, 11 min	Outer ring
	Bearing 2_2	2 h, 41 min	Outer ring
	Bearing 2_3	8 h, 53 min	Cage
	Bearing 2_4	42 min	Outer ring
	Bearing 2_5	5 h, 39 min	Outer ring
3	Bearing 3_1	42 h, 18 min	Inner ring and cage
	Bearing 3_2	41 h, 36 min	Outer ring
	Bearing 3_3	6 h, 11 min	Inner ring
	Bearing 3_4	25 h, 15 min	Inner ring
	Bearing 3_5	1 h, 54 min	Outer ring

input data are independent. Moreover, Li et al. [34–38] proposed a supervised deep learning direct convolutional neural network (CNN) with the sliding windows technique to map the relationships between the turbofan engine dataset and measured RUL values. Although direct CNN variants can manage high-dimensional data, they cannot capture the time-series trends of long-term pre-processed data. To improve the accuracy of RUL estimations made by direct CNN models, a supervised learning method, the direct long short-term memory (LSTM), was proposed, along with other methods, to learn the relationships that are able to capture the long-term degradation trends of turbofan engines [39–45]. The LSTM considers RUL prediction to be a supervised learning process, during which the inputs are sensor signals truncated by the sliding windows technique and the outputs are the corresponding RUL values, as illustrated in Fig. 9.

3.1.2 Indirect mapping methods

Indirect mapping methods involve extracting representative temporal and spatial features, referred to as health indicators (HIs), when the features are one-dimensional and strongly correlated with the RUL. Subsequently, these.

extracted features are fed into a predictor, as discussed in Section 3.1.1, to establish mapping relationships between the extracted features and the corresponding RUL values. The feature extraction processes in indirect mapping methods were critically reviewed and compared during this study because feature extraction is vital. Statistical models and neural network models typically extract representative features from turbofan engine datasets. Statistical models, such as the linear regression model proposed by Maulana

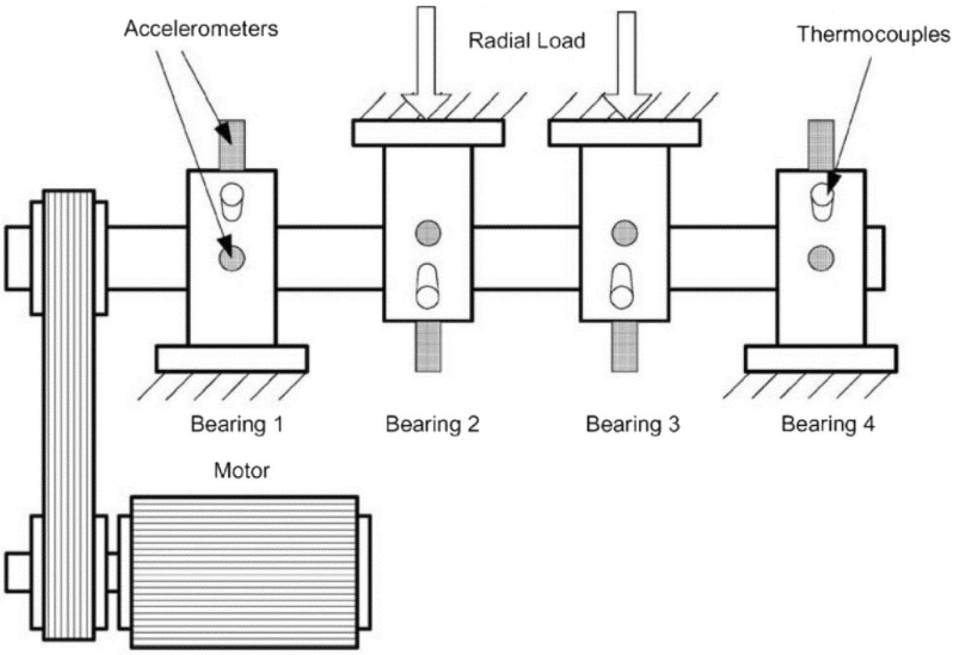


Fig. 8 Accelerated bearing degradation test platform [28].

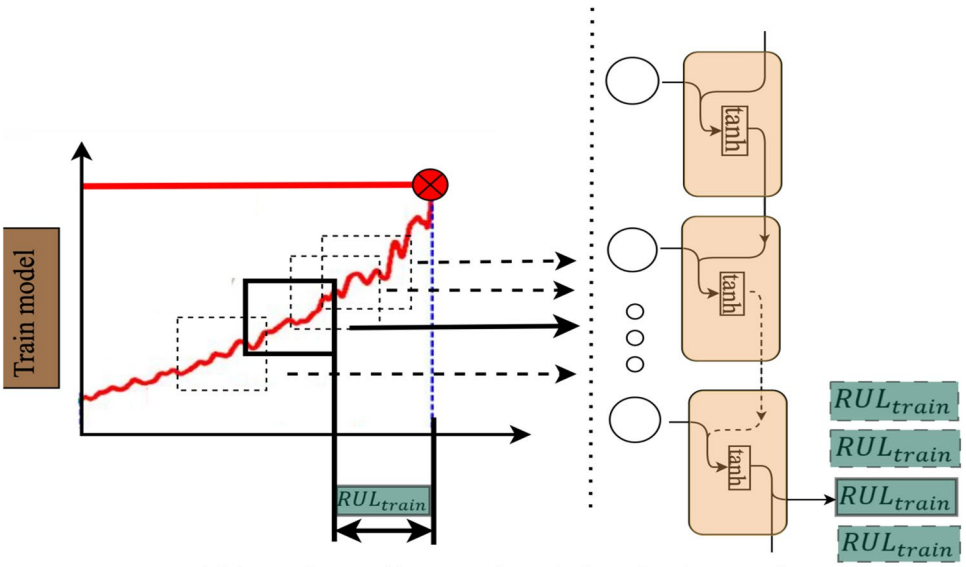


Fig. 9 Schematic diagram of the process of using the direct LSTM for RUL prediction.

et al. [46, 47], create a multivariate linear function to visualize the degradation trends using one-dimensional representative spatial features. However, the degradation trends of turbofan engines possess substantial nonlinear variability; thus, it is challenging to adequately describe them using linear functions. Furthermore, Fu et al. [48, 49] proposed the piecewise linear model and Bi et al. [50] offered the hidden Markov model (HMM) as way to fit the degradation trends. The piecewise linear model and the HMM can extract more complex degradation trends. However, sole reliance on one-dimensional representative spatial features may not produce full representations of the degradation trends, especially for complex and sophisticated turbofan engines. Therefore, multi-dimensional features are extracted from the pre-processed data to capture difficult degradation trends. For instance, Tang et al. [51] proposed an RF algorithm to measure feature significance, thereby aiding the selection of the most relevant spatial features. In addition, principal component analysis (PCA) can also be used to extract uncorrelated multi-dimensional spatial features using orthogonal transformation [52–57]. Although these methods extract multi-dimensional features from pre-processed data, it is important to note that some information may be lost if incorrect principal components are selected [58].

Moreover, deep learning is the leading technology used to automatically extract multi-dimensional features from pre-processed data. Ellefsen et al. [59] and Zhang et al. [60] proposed a deep belief network (DBN) to automatically extract spatial features from pre-processed data. In addition, a CNN and a CNN variant were proposed to extract spatial features from the pre-processed data using a convolutional layer [24, 61–67]. Although CNNs can extract features more effectively than DBNs, CNNs cannot extract temporal features. An recurrent neural network (RNN), which is a classical network used for extracting temporal features, was proposed by Zhao et al. [68] to predict the RUL. However, gradient explosion and vanishing can occur when processing long time-series data. A new RNN variant, the long short-term memory (LSTM) network, was introduced to address this issue. The LSTM network includes a memory cell, which was further enhanced by integrating an attention mechanism to weigh the extracted features [69]. However, the LSTM network requires a large number of data for training since it must be integrated with the predictor and trained as a whole. The unsupervised model called the Vaillant autoencoder relaxes this data requirement by extracting features individually [70]. The representative spatial features, Z_t, include all the degradation information extracted by the Vaillant autoencoder when the reconstructed data is similar to the original data, as shown in Fig. 10. However, the Vaillant autoencoder cannot extract temporal features. To enhance the temporal-feature extraction using the Vaillant autoencoder, de Beaulieu et al. [71, 72] combined the LSTM network with the autoencoder to propose the LSTM encoder–decoder. The LSTM encoder–decoder utilizes LSTM architecture for both the encoding and decoding processes and can therefore extract high-quality representative temporal and spatial features. A summary of the feature extraction techniques discussed above is presented in Table 4.

3.1.3 Performance assessment

RUL estimation methods are evaluated based on prediction accuracy and computational cost. In light of the critical review of RUL estimation methods earlier in this study, the accuracy of these models is now under assessment. The accuracy metrics used for the RUL estimation assessment include the mean absolute error (MAE), the score function, the relative error, the cumulative relative accuracy (CRA), and the root mean square error (RMSE) [93–95]. The definitions of these metrics are presented as.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|Y_{real} - Y_{pre}| \tag{1}$$

$$Score = \begin{cases} \sum_{i=1}^{n}\exp\left(-\frac{Y_{pre}-Y_{real}}{13}\right) - 1 if(Y_{pre} - Y_{real} \le 0) \\ \sum_{i=1}^{n}\exp\left(-\frac{Y_{pre}-Y_{real}}{10}\right) - 1 if(Y_{pre} - Y_{real} > 0) \end{cases} \tag{2}$$

$$Error = \left|\frac{Y_{pre} - Y_{real}}{Y_{real}}\right| \times 100\% \tag{3}$$

$$CRA = \frac{1}{n}\sum_{n=1}^{n}(1 - \frac{|Y_{real} - Y_{pre}|}{Y_{real}}) \tag{4}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_{real} - Y_{pre})^2} \tag{5}$$

where n represents the number of data points or samples, Y_{real} is the actual RUL value of the ith sample, and Y_{pre} is the predicted RUL value of the ith sample.

Wu et al. [23] and other researchers utilize RMSE as a primary metric to assess the deviations between predicted and actual RUL values. RMSE's advantage lies in sharing the same unit of measurement as the target variable, in contrast to metrics like Score, Error, and CRA. In addition, RMSE places more emphasis on large errors compared to MAE, thus penalizing substantial deviations more harshly. Given the crucial nature of precise RUL predictions, RMSE stands out as the preferred metric in such evaluations. It is noteworthy that NASA released two turbofan engine datasets, C-MAPSS and N-CMAPSS. However, it is essential to acknowledge that utilizing the N-CMAPSS dataset to assess performance may yield less robust results than using the C-MAPSS dataset because it is relatively recent and needs more studies to support comprehensive performance evaluations. Therefore, RMSE scores were obtained for all the RUL estimation models used in the C-MAPSS dataset, as shown in Fig. 11. It is vital to note that some of the methods discussed earlier in this study, for which RMSE scores were not presented, are not included in Fig. 11.

The first three methods in Fig. 11, which include the PF, direct CNN, and LSTM methods (in this case, the CNN and LSTM methods can directly predict the RUL without feature extraction), are direct mapping methods that do not require the extraction of representative features. By contrast, the last four methods, which include the autoencoder, CNN, and LSTM methods (in this case, the CNN and LSTM methods are used as feature extractors), are indirect mapping methods. These feature extraction methods are often coupled with regressors to predict the RUL. Figure 11 below illustrates that the RMSE scores for RUL estimation obtained by the indirect mapping methods were lower than those obtained by the direct mapping methods, which underscores the significance of feature extraction in the accurate estimation of RUL. Furthermore, it was observed that the lowest RMSE values were obtained when the autoencoder or autoencoder variant models were used to extract features.

Moreover, the computation cost and whether is easy to implement is also vital metric for choosing the RUL

Table 4 Summary of the feature extraction techniques employed in indirect mapping methods for turbofan engine RUL estimation

	Feature extraction method	Feature extraction technique	Advantages	Limitations	References
Statistical models	One-dimensional feature extraction	Linear regression	Can visualize the degradation trend	The one-dimensional features cannot represent the degradation trend	[46, 47]
		Piecewise linear model			[48, 49]
		HMM			[50]
	Multi-dimensional feature extraction	RF algorithm	Can extract abstract features	Cannot capture nonlinear relationships between the features	[51]
		PCA			[52–55]
		Kernel PCA	Able to capture the nonlinear relationships between the features	Some information may be lost if the wrong number of principal components is chosen	[56]
		Probabilistic PCA			[57]
Neural network models	DBN	Restricted Boltzmann machine (RBM)	Can learn abstract features from raw unlabeled data	Inefficient	[73]
		DBN			[60]
	CNN	CNN	Able to automatically extract features	Cannot extract long-term temporal features	[24, 61, 62]
		Attention-based CNN			[63, 74]
		CNN-based spatiotemporal attention mechanism	Can extract both spatial and temporal features		[65]
		Multi-scale residual temporal convolutional network (MSR-TCN)			[66]
		Graph convolutional network	Can extract irregular and complex graph features		[67]
	RNN	Enhanced recurrent neural network	Can extract temporal features	Gradient explosion and vanishing occur	[68]
	LSTM	Attention-based Bi-LSTM	Able to avoid gradient explosion and vanishing	Many data are required for training	[69]
		Attention-based LSTM			[76–79]
		Temporal attention-based LSTM			[64]
		Multicellular LSTM			[80]
		Combination of LSTM, Bi-LSTM, and CNN	Combines three deep learning networks to extract deep features automatically	The model can be easily overfitted during training	[81]
		Combination of CNN and LSTM			[82]
	Autoencoder	Vaillant autoencoder	Can perform unsupervised feature extraction	Cannot extract temporal features	[70]
		Improved stack sparse autoencoder			[83]
		Variational autoencoder			[71, 84–86]
		Spare autoencoder			[87]
		Denoising autoencoder			[88]
		CNN autoencoder			[89]
		Time-series memory autoencoder	Can perform unsupervised extraction of temporal features	Cannot extract ultra-long temporal features	[90]
		Vaillant encoder–decoder	Can perform supervised feature extraction	Cannot extract spatial features	[91]
		RNN encoder–decoder	Can perform supervised extraction of temporal features		[92]
		LSTM encoder–decoder			[71, 72]

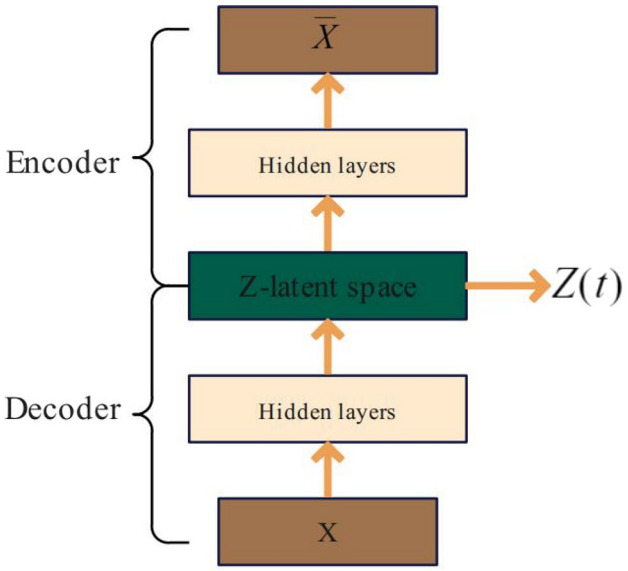


Fig. 10 Schematic diagram of the Vaillant autoencoder architecture.

estimation model. Some models require extensive computational resources and are challenging to implement in physical setups. The direct mapping method, which predicts RUL values without the need for feature extraction, is efficient and straightforward to implement. This method comprises statistical models like direct CNN and direct LSTM, offering low time consumption and ease of implementation in real-world scenarios. Conversely, the indirect mapping method involves feature extraction and predictor, leading to higher time consumption and complexity in implementation due to the additional steps involved.

3.2 Bearings

The XJTU-SY, University of Cincinnati, and IEEE 2012 PHM bearing datasets are online datasets that are commonly used for RUL estimation, as described in Section 2. Both datasets include vibration and temperature signals. However, these signal features contain limited degradation information; thus, there is a need to extract representative temporal and spatial features before feeding them into the predictor. Therefore, the indirect mapping methods for bearing RUL estimation are critically reviewed and assessed below.

3.2.1 Indirect mapping methods

As for turbofan engine RUL estimation, feature extraction techniques are also vital for bearing RUL estimation. Therefore, the feature extraction techniques were critically reviewed during this study. Statistical and neural network models typically extract representative features from bearing datasets. The HI construction statistical model was proposed by Ahmad et al. [96]. In this model, the HI at time t can be calculated using Eq. 6

$$HI(t) = \frac{x_{RMS}(t)}{x_{RMS}(\Delta t_{baseline})} \tag{6}$$

where $x_{RMS}(t)$ represents the root mean square (RMS) of the vibration signal at t and $x_{RMS}(\Delta t_{baseline})$ is the RMS of the vibration signal for the baseline condition. Then, a linear rectification technique (LRT) is applied to smooth the HI curve. However, the constructed HI lacks the frequency and time–frequency domain information of the vibration signal.

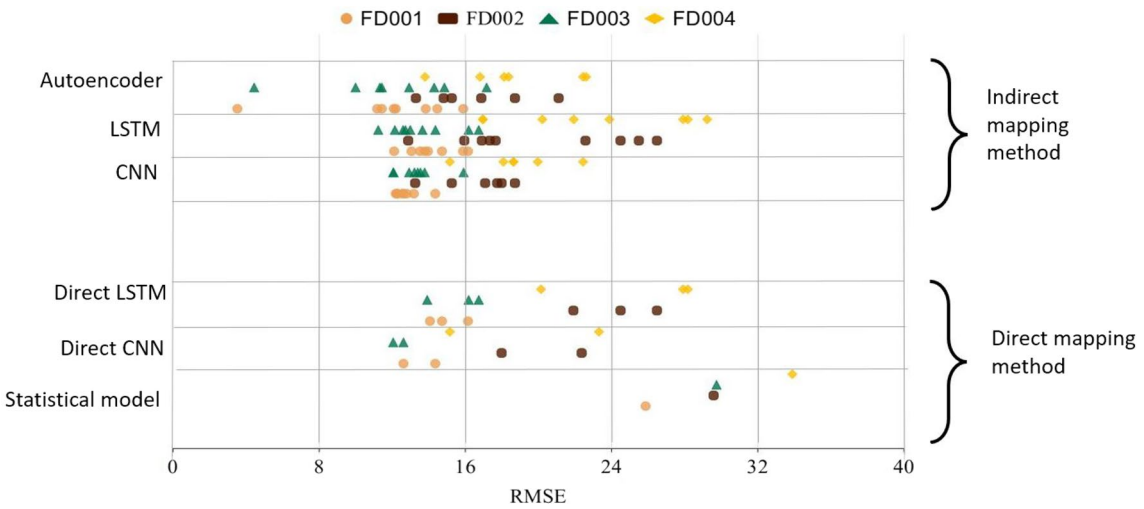


Fig. 11 RMSE scores for the models used with the C-MAPSS dataset.

Moreover, data transformation techniques, such as the fast Fourier transform (FFT) and wavelet transform techniques, are used to change the vibration signals to frequency and time–frequency domain data, as presented in [97–100]. Examples of time-domain features include the peak, peak-to-peak, skewness, kurtosis, and waveform indicator values. Frequency-domain features include the spectrogram, peak frequency, frequency bandwidth, and spectral indicators, whereas the time–frequency domain features consist of the wavelet packet energy spectra. It is worth noting that different types of features are extracted using these different data transformation techniques, of which not all exhibit significant degradation information. To select relevant features, Zhao et al. [101] proposed a fitting-curve derivative method for the maximum frequency spectrum density to choose relevant features. In addition, PCA, which is a dimensionality reduction technique that involves finding a set of orthogonal axes along which the data exhibit the largest variations, was proposed by Liu et al. [102]. However, these methods cannot extract complex and nonlinear features from bearing datasets.

To extract complex and nonlinear features, Zeng et al. [94, 103] proposed a neural network model called CNN to extract complex features from bearing datasets. Although this method can automatically extract complex and nonlinear features from pre-processed data, it performs poorly when extracting temporal features. To address this problem, Cao et al. [104] proposed a sophisticated CNN variant called a temporal convolution network (TCN) that utilizes causal and dilated convolutions, as well as residual connections, to enhance temporal feature extraction. In addition, the LSTM was proposed by Li et al. [29] to extract temporal features. However, the TCN and LSTM both require a large number of data points for training since they must be integrated

Table 5 Summary of the feature extraction techniques employed for bearing RUL estimation

	Feature extraction method	Feature extraction technique	Advantages	Limitations	References
Statistical models	HI construction	Gaussian mixture model and Kullback–Leibler divergence	Can visualize the HI of the machinery	Lacks frequency and time–frequency domain features	[109]
		HI construction through formulation			[96]
	Data transformation	Fourier transform	Possess simple calculation formulas	Cannot extract complex or nonlinear features	[97–99, 110]
		Improved empirical wavelet transform (IEWT)			[100]
		Wavelet packet transform (WPT)			[111]
		Envelope analysis of frequency-domain data			[112]
		Fitting-curve derivative method for the maximum power spectrum density			[101]
		Mean impact value			[113]
		Spearman coefficient			[114]
Neural Networks	DBN	Restricted Boltzmann machine (RBM)	Can extract complex and nonlinear features	Inefficient	[115]
	Deep neural network	DNN		Cannot extract temporal features	[116]
	CNN	CNN	Can automatically extract features		[94, 103]
		Spatial pyramid pooling			[117]
		CNN with polynomial decaying learning rate			[118]
		Temporal convolutional network	Can extract temporal features	A large number of data points are required for training	[104]
	LSTM	2D-LSTM fusion network			[29]
	Autoencoder	Vaillant autoencoder	Can perform unsupervised feature extraction	Cannot extract temporal features	[107]
		Convolutional autoencoder			[105, 108]
		Transfer stacked contractive autoencoder			[106]

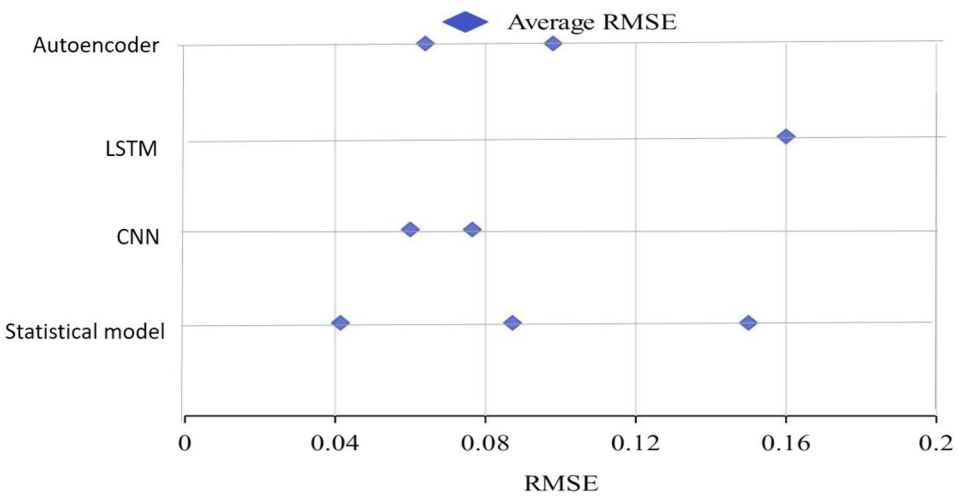


Fig. 12 RMSE scores for all the models used with the IEEE 2012 PHM bearing dataset.

with a predictor and trained as a whole. The unsupervised model, the Vaillant autoencoder, and the autoencoder variants reduce the data requirements by extracting features individually [105–108]. A summary of the feature extraction techniques discussed above is presented in Table 5.

3.2.2 Performance assessment

An assessment of the accuracy of the RUL estimation models, which were reviewed earlier with regard to theoretical analysis, is presented next. The IEEE 2012 PHM bearing dataset is widely recognized as a classic and extensively used online bearing dataset. Therefore, various RUL predictive models were evaluated using this dataset. When considering the multiple cases of bearing run-to-failure included in the online dataset, the average RMSE was selected as the evaluation metric used in the result comparison. The average RMSE is defined in Eq. 7:

$$Average\ RMSE = \sum_{i=1}^{N} \frac{1}{N} RMSE_i \tag{7}$$

where N is the number of samples and i is the number of the current sample.

The RMSE scores for all the models used with the IEEE 2012 PHM bearing dataset are depicted in Fig. 12, in which the models that employ CNN and autoencoder techniques for feature extraction exhibited superior prediction accuracies.

4 Conclusions

This paper presents a critical review and an assessment of ML-based RUL estimation models by separating analyses of each step in the RUL prediction process and categorising similar models for simple comparisons. The C-MAPSS and N-CMAPSS online turbofan engine datasets, as well as the IEEE 2012 PHM, XJTU-SY, and University of Cincinnati online bearing datasets, were selected as the objectives. One hundred ten high-quality relevant articles that discuss RUL prediction in the context of these datasets were critically reviewed. First, the data analysis and pre-processing were reviewed for each dataset. Second, turbofan engine and bearing RUL estimation models were separately reviewed and evaluated. Direct and indirect mapping methods are the primary methods used for turbofan engine RUL prediction. The results indicate that the LSTM model is suitable for managing time-series data within a direct mapping method. In addition, an autoencoder variant known as the LSTM encoder–decoder demonstrated the ability to extract temporal and spatial features in an unsupervised learning environment within an indirect mapping method. Furthermore, the RMSE values obtained with the indirect mapping method are lower than those obtained with the direct mapping method, and the autoencoder and its variant produced relatively low RMSE scores. When the CNN and autoencoder are used to extract features, they can produce the lowest RMSE values for bearing RUL estimation.

The findings of this research suggest several areas of future research that have the potential to advance RUL prediction for turbofan engines and bearings:

- The exploration of alternative deep learning methods not yet applied to RUL prediction, such as reinforcement learning, graph neural networks (GNN), and transformers.
- The development of a generalized or large-scale model capable of predicting the RULs of various objects, including turbofan engines and bearings.
- Improvements in the early-stage RUL prediction accuracy.

- The investigation of failure thresholds for different working conditions.
- The application of reinforcement learning techniques to address limited sample sizes and enhance model robustness and generalization.
- The development of hyperparameter optimization algorithm for deep learning model to enhance quality of extracted features.

Declarations

Ethics approval This is not applicable because this article does not contain any studies with human or animal subjects.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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